

CHEMISTRY | UNIT 02 | LESSON 03

Isotopes, Atomic Mass, and Atomic Identity

Notebook Evidence - Independent Application

Student name:		Date:	
MLA standard:	MLA.CHEM.ATO.02	Submission:	Notebook Evidence - Independent Application

Directions

Open PhET Isotopes and Atomic Mass from Moodle. Use the Isotopes screen and the Mix Isotopes screen. Record the simulation evidence below. Submit this PDF in U2 L3 Notebook Evidence Submission.

Part A - Isotope evidence

Trial	Element	p+	n0	Mass #	Stable?
1. carbon-12					
2. carbon-14					
3. neon-20					
4. neon-22					

Part B - Mix isotopes

Create a sample containing 90% neon-20 and 10% neon-22. Record the values and calculate the weighted average.

Isotope	Fraction	Mass contribution
neon-20	0.90	
neon-22	0.10	
Total		

Show the calculation and explain why the average is closer to 20 than 22.
